

PRACTICAL ASSESSMENT

PART B — Scenario-Based Practical Evaluation

Ubuntu Linux | Windows Server 2019 | Windows PowerShell

Candidate Name: _____	Date: _____
Roll No: _____	Scenario Attempted: Scenario No. _____
Total Marks: 80 (Attempt ANY ONE Scenario Only)	

Scenario 1: Ubuntu Linux — User Management, File System Security and Shell Automation	
Topic: Ubuntu Linux Administration	Total Marks 80 <i>Intermediate to Hard</i>
Context: You are a Linux system administrator for a growing startup. A new operations engineer 'sysops' must be onboarded with a secure account, a structured workspace must be built under their home directory with correct permissions, log analysis must be performed using command pipelines, and a disk-monitoring automation script must be deployed.	Attempt This Scenario ONLY if you choose it
Objective: Create and configure a user account with sudo access, build a nested directory structure, apply octal permissions and recursive ownership, construct text-processing pipelines, write a disk-monitoring bash script, perform a system audit using find and ps, and install a networking utility.	

Tasks

#	Task Description	Marks
1	Open the terminal. Create a new user account named 'sysops' with a home directory and bash shell using the correct useradd flags (-m and -s /bin/bash). Set a password using passwd. Verify the account was created by running: cat /etc/passwd grep sysops Confirm the entry is present and show the full line to the examiner.	6
2	Add 'sysops' to the sudo group using usermod -aG sudo sysops. Verify group membership by running 'id sysops' and confirming sudo appears in the groups output. Run 'groups sysops' as a second confirmation. Show both command outputs side-by-side to the examiner.	8
3	Create the following nested directory structure under /home/sysops using a single mkdir -p command: workspace/api, workspace/frontend, reports Verify the structure with 'ls -lR /home/sysops' and confirm all three subdirectories exist before proceeding.	8
4	Apply permissions using chmod with octal notation: /home/sysops/workspace = 755 workspace/api = 750 workspace/frontend = 750 reports = 700	10

	Change ownership of /home/sysops recursively to sysops:sysops using chown -R. Verify all permissions and ownership with: <pre>ls -la /home/sysops/ ls -la /home/sysops/workspace/</pre>	
5	Navigate to /var/log. Construct a single pipeline command that: (1) uses grep -i to find 'warning' in auth.log (2) pipes to sort (3) pipes to uniq to remove duplicate lines (4) redirects output using >> to /home/sysops/reports/warning_log.txt Display the file using cat and count total lines using wc -l. Show both outputs to the examiner.	12
6	Create /home/sysops/disk_monitor.sh with: (a) correct shebang line #!/bin/bash (b) a variable LOGFILE='/home/sysops/reports/diskcheck_\$(date +%Y%m%d).txt' (c) write hostname, current user, and current date to the file (d) append disk usage using df -h (e) append memory usage using free -m (f) append top 5 memory-consuming processes using ps aux --sort=-%mem head -6 Set permissions to 755 with chmod, execute the script, and verify the report file contains all expected sections.	14
7	Use find to locate all files under /home/sysops with permissions set to 700: <pre>find /home/sysops -perm 700</pre> Also list all running processes owned by sysops: <pre>ps aux grep sysops</pre> Append both command outputs to /home/sysops/reports/system_audit.txt using >>. Display the final file with cat to the examiner. Note: An empty grep result for processes is acceptable — document it as-is.	10
8	Run sudo apt update to refresh package repositories. Install the 'net-tools' package using sudo apt install net-tools -y. After installation run ifconfig to confirm the tool is working. Verify installation with: <pre>dpkg -l grep net-tools</pre> Append the package name and installed version to /home/sysops/reports/installed_packages.txt using echo and >>.	12
TOTAL MARKS		80

Examiner Remarks: 	Marks Obtained: _____ / 80
-----------------------------------	---

Scenario 2: Windows Server 2019 — ADDS Setup, DC Promotion and Domain Join	
Topic: Windows Server 2019 — Active Directory Domain Services	
Total Marks 80 <i>Intermediate to Hard</i>	
Context: A fresh Windows Server 2019 instance has been provisioned in VirtualBox for your organisation's training lab. You must configure server identity and assign a static IP address, install and configure ADDS, promote the server to a Domain Controller for techlab.local, join a Windows 10 workstation to the new domain, and verify end-to-end functionality with a test domain user login.	Attempt This Scenario ONLY if you choose it
Objective: Configure server hostname and static IP, install the ADDS role, promote to Domain Controller, validate domain structure in ADUC, configure client networking, perform domain join from a Windows 10 machine, and verify end-to-end domain authentication.	

Tasks

#	Task Description	Marks
1	On Windows Server 2019, open Server Manager > Local Server. Click the computer name link and rename the server to LABDC01. Then open Network Connections, select the Ethernet adapter, open IPv4 Properties and configure: IP Address = 10.0.0.50 Subnet Mask = 255.255.255.0 Default Gateway = 10.0.0.1 Preferred DNS = 10.0.0.50 Apply settings and restart the server if prompted before continuing.	6
2	In Server Manager click Add Roles and Features. Proceed through the wizard and select Active Directory Domain Services from the Server Roles list. Accept all required additional features when prompted. Complete the installation and wait for the yellow notification flag to appear in Server Manager's top bar before proceeding to the next task.	8
3	Click the notification flag in Server Manager and select Promote this server to a domain controller. In the deployment wizard: - Select Add a new forest - Set Root domain name = techlab.local - On the Domain Controller Options page keep all defaults - Set a DSRM password of your choice Complete the prerequisites check and click Install. The server will restart automatically upon completion.	10

4	After restart, log in as TECHLAB\Administrator. Open ADUC from Server Manager > Tools > Active Directory Users and Computers. Verify that techlab.local is visible with all default containers: Builtin, Computers, Domain Controllers, Users. Confirm LABDC01 appears inside the Domain Controllers OU. Take a screenshot or show this view to the examiner before proceeding.	10
5	On the Windows 10 client machine, open Ethernet adapter IPv4 Properties and configure: IP Address = 10.0.0.75 Subnet Mask = 255.255.255.0 Default Gateway = 10.0.0.1 Preferred DNS = 10.0.0.50 Open Command Prompt and run all three verification commands: (a) ipconfig — confirm the correct IP settings (b) ping 10.0.0.50 — confirm 0% packet loss (c) nslookup labdc01.techlab.local — confirm DNS resolution All three must succeed before proceeding to Task 6.	12
6	On Windows 10, verify local administrator rights via Computer Management > Local Users and Groups. Then join the domain: System Properties > Change Settings > Computer Name tab > Change Select Domain and type: techlab.local > OK Enter credentials: Administrator / (server password) Accept the welcome message. Restart the Windows 10 machine when prompted.	14
7	After the Windows 10 machine restarts, log in as TECHLAB\Administrator. Open System Properties and confirm the Domain field displays techlab.local. On LABDC01, open ADUC > Computers OU and confirm the Windows 10 computer object has appeared in the list, verifying successful domain registration. Show both the client System Properties and the ADUC Computers OU to the examiner.	10
8	On LABDC01 in ADUC, create a test domain user inside the Users OU: First Name = Demo Last Name = Account Logon Name = demo.account Password = Demo@5678 Options = Password never expires (checked) Log out of Windows 10 and log back in as TECHLAB\demo.account with the password above. Confirm you reach the Windows 10 desktop successfully — this proves end-to-end domain authentication is working.	10

TOTAL MARKS		80
------------------------	--	-----------

Examiner Remarks: _____	Marks Obtained: _____ / 80
-----------------------------------	---------------------------------------